

Observable extrapolations for two-body resonances via eigenvector continuation

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Abstract

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I. INTRODUCTION

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II. COMPLEX-SCALING EIGENVECTOR CONTINUATION

Complex-scaling eigenvector continuation (CSEC).

A. Complex scaling

A resonance is defined as a pole of the S matrix (equivalently, a zero of the Jost function) at a complex energy on the second Riemann sheet; i.e. $p = p_r - i\gamma$ ($k, \gamma > 0$). At any zero of the Jost function p_0 , $\phi_{p_0}(r) \propto f_{p_0}^+(r)$, where $\phi_{p_0}(r)$ is the regular solution of the radial equation and $f_{p_0}^+(r)$ is the outgoing Jost function, defined by a solution of the radial equation with boundary conditions $f_{p_0}^+(r) \xrightarrow{r \rightarrow \infty} h^+(p_0 r)$ (where $h^+(r) \sim e^{ir}$ is the outgoing Ricatti-Hankel function). Because $\gamma > 0$, $\phi_p(r) \propto h^+(pr) \sim e^{ip_r r} e^{\gamma r}$ ($r \rightarrow \infty$) diverges, and is not L^2 under the Hilbert space norm.

Complex scaling is defined by the unitary operation U_θ , defined such that

$$\langle \mathbf{r} | U_\theta | \Psi \rangle = e^{id/2} \langle \mathbf{r} e^{i\theta} | \Psi \rangle \sim U_\theta \Psi(\mathbf{r}) = e^{id/2} \Psi(e^{i\theta} \mathbf{r}),$$

where d is the dimension of the Hilbert space, $|\Psi\rangle$ is an eigenfunction of the radial equation (in the rigged Hilbert space sense), and generally $\theta \in \mathbb{R}$. Under this operation,

$$\phi_p(r) \rightarrow U_\theta \phi_p(r) = e^{i\theta/2} \phi_p(re^{i\theta}) \xrightarrow{r \rightarrow \infty} e^{i\theta/2} e^{ir(p_r \cos \theta + \gamma \sin \theta)} e^{-r(p_r \sin \theta - \gamma \cos \theta)}.$$

The divergence is tamed as long as $p_r \sin \theta > \gamma \cos \theta \rightarrow \theta > \arg(k)$ or $\theta > \arg(E)/2$.

Theorem II.1. *Aguilar-Balslev-Combes (ABC) Let $H = H_0 + V$, $H = -\frac{\hbar^2}{2m} \Delta$ on $L^2(\mathbb{R}^d)$ and suppose V is sufficiently well behaved under analytic dilation (**TODO: TIGHTEN THIS UP; SUFFICIENTLY WELL-BEHAVED HOW?**)*

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B. Eigenvector continuation

C. Continuation to complex rotation angle

III. APPLICATIONS

A. Simple two-body systems

1. Narrow s-wave resonance

2. Narrow p-wave resonance

3. Wide p-wave resonance

B. Three-boson system

IV. UNCERTAINTY ESTIMATION

V. DISCUSSION AND OUTLOOK

VI. NOTES

VII. TODO

<https://arxiv.org/abs/1007.4319> (extension of ABC theorem to a more abstract type of manifold)

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